Total Printed Pages: 02

Roll No. :.....

ECA101

Examination –MAY 2024

B.Tech. I Sem : Electronics and Communication

Basic Electronics

Time : 2 Hrs

Max. Marks : 60

Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

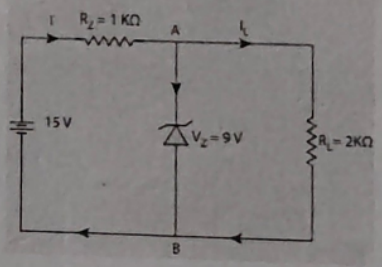
Word limit be observed as follows:

Part a & b – Objective type/Fill in the blanks/One sentence answer.

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)	Which semiconductor material is mostly preferred to manufacture electronic devices?	02	1,2	CO1
	(b)	For tunning circuit which type of diode is used? Justify your answer.	02	3,6	CO4
	(c)	Compare Zener and avalanche breakdown	03	3,4	CO2
		OR			
		Draw the VI characteristics of tunnel diode.	03	3,4	CO2
	(d)	With the help of VI Characteristics explain the working of Zener diode.	05	3,4	CO3
		OR			
		Find the current through the Zener diode when the load resistance is 1 KΩ. Use diode approximation.	05	3,6	CO4



Q.2	(a)	What are the three types of transistor configurations?	02	1,2	CO1
	(b)	Why emitter is heavily doped?	02	3,4	CO2
	(c)	Explain punch through.	03	3,6	CO3
		OR			
		Derive the relation between alpha and beta.	03	3,4	CO2
	(d)	Sketch the Circuit diagram of a NPN common base configuration explain its working and VI characteristics	05	3,4	CO2
		OR			

		Sketch the Circuit diagram of a NPN common emitter configuration explain its working and VI characteristics	05	3,4	CO3
Q.3	(a)	Give the advantage of h-parameter analysis.	02	1,2	CO1
	(b)	The dimension of hie parameter are i)mho ii)ohm iii)Farad iv) none	02	1,2	CO1
	(c)	Draw the hybrid equivalent circuit for common base transistor.	03	3,4	CO2
		OR			
		Draw the hybrid equivalent circuit for common emitter transistor.	03	3,4	CO2
	(d)	Derive the equations for current gain and voltage gain of a CE amplifier using h-parameters.	05	3,4	CO2
		OR			
		Give low frequency h-parameter model of BJT and explain the same	05	3,6	CO3
Q.4	(a)	Name different types of coupling .	02	1,2	CO1
	(b)	Define multistage Amplifier .	02	1,2	CO1
	(c)	Discuss Darlington amplifier .	03	3,4	CO2
		OR			
		Differentiate between cascade and cascode amplifier .	03	3,4	CO2
	(d)	A multistage amplifier employs five stages each of which has a power gain of 30. What is the total gain of the amplifier in db .	05	3,4	CO2
		OR			
		With the help of neat sketches explain RC coupled amplifier.	05	3,4	CO2
Q.5	(a)	A tuned amplifier is used in ..... application i)Radio frequency ii)Low frequency iii) Audio frequency iv) None of the above	02	1,2	CO1
	(b)	Write the advantages of tuned amplifier?	02	1,2	CO1
	(c)	Discuss harmonic Distortion	03	3,4	CO2
		OR			
		Differentiate A, B, AB and C type of power amplifier.	03	3,6	CO3
	(d)	Discuss the characteristics and response of single tuned amplifiers.	05	3,4	CO2
		OR			
		Discuss the characteristics and response of staggered tuned amplifiers.	05	3,4	CO2

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

*****